

## SAFETY DATA SHEET

Revision Date 16-December-2019

Revision Number 1

### 1. Identification

**Product Name** 4'-Fluorobiphenyl-4-sulfonyl chloride, 97%

**Cat No. :** AC450320000, AC450320010, AC450320050

**CAS-No** 116748-66-4  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.  
**Details of the supplier of the safety data sheet**

#### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

#### **Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01  
/ **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 /  
**Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 /  
**Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                  |              |                           |
|------------------------------------------------------------------|--------------|---------------------------|
| <b>Acute Inhalation Toxicity</b>                                 | Category 2   | (based on evolved HF gas) |
| <b>Skin Corrosion/Irritation</b>                                 | Category 1 B |                           |
| <b>Serious Eye Damage/Eye Irritation</b>                         | Category 1   |                           |
| <b>Specific target organ toxicity (single exposure)</b>          | Category 3   |                           |
| Target Organs - Respiratory system.                              |              |                           |
| <b>Health Hazards Not Otherwise Classified</b>                   | Category 1   |                           |
| In contact with water, releases gases which are toxic if inhaled |              |                           |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Fatal if inhaled  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
In contact with water, releases gases which are toxic if inhaled

**Precautionary Statements****Prevention**

Do not allow contact with water  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Use only outdoors or in a well-ventilated area  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                                    | CAS-No      | Weight % |
|----------------------------------------------|-------------|----------|
| 4'-Fluoro(1,1'-biphenyl)-4-sulfonyl chloride | 116748-66-4 | >95      |

### 4. First-aid measures

|                       |                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b> | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                            |
| <b>Eye Contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.                                                                                                                                                                          |
| <b>Skin Contact</b>   | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.                                                                                                                                                                                                          |
| <b>Inhalation</b>     | Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |
| <b>Ingestion</b>      | Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water.                                                                                                                                                                                                                                                      |

Never give anything by mouth to an unconscious person.

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | No information available                                          |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | No information available                                          |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

Contact with water liberates toxic gas. The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Hydrogen chloride gas Carbon monoxide (CO) Carbon dioxide (CO<sub>2</sub>) Chlorine Hydrogen fluoride Sulfur oxides Thermal decomposition can lead to release of irritating gases and vapors

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 0                   | 0                  | W                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.                                  |
| <b>Environmental Precautions</b>            | Should not be released into the environment. Do not allow material to contaminate ground water system. See Section 12 for additional Ecological Information. |
| <b>Methods for Containment and Clean Up</b> | Sweep up or vacuum up spillage and collect in suitable container for disposal. Avoid dust formation. Do not expose spill to water.                           |

## 7. Handling and storage

|                 |                                                                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/face protection. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. Do not allow contact with water. |
| <b>Storage</b>  | Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. To maintain product quality. Store under an inert atmosphere. Keep away from water or moist air.                 |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**
**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Butyl rubber   | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                  |                               |
|----------------------------------|-------------------------------|
| Physical State                   | Solid                         |
| Appearance                       | White - Off-white             |
| Odor                             | No information available      |
| Odor Threshold                   | No information available      |
| pH                               | No information available      |
| Melting Point/Range              | 82 - 84 °C / 179.6 - 183.2 °F |
| Boiling Point/Range              | No information available      |
| Flash Point                      | No information available      |
| Evaporation Rate                 | Not applicable                |
| Flammability (solid,gas)         | No information available      |
| Flammability or explosive limits |                               |
| Upper                            | No data available             |
| Lower                            | No data available             |
| Vapor Pressure                   | No information available      |
| Vapor Density                    | Not applicable                |

|                                        |                          |
|----------------------------------------|--------------------------|
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | C12 H8 Cl F O2 S         |
| Molecular Weight                       | 270.71                   |

## 10. Stability and reactivity

|                                  |                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                                                                  |
| Stability                        | Stable under recommended storage conditions. Contact with water liberates toxic gas.                                                                                                                 |
| Conditions to Avoid              | Exposure to moist air or water. Incompatible products. Excess heat.                                                                                                                                  |
| Incompatible Materials           | Water, Bases, Alcohols, Amines, Strong oxidizing agents                                                                                                                                              |
| Hazardous Decomposition Products | Hydrogen chloride gas, Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Chlorine, Hydrogen fluoride, Sulfur oxides, Thermal decomposition can lead to release of irritating gases and vapors |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                                                             |
| Hazardous Reactions              | None under normal processing.                                                                                                                                                                        |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information** No information available

**Toxicologically Synergistic** No information available

**Products**

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** Causes burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                    | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------------------|-------------|------------|------------|------------|------------|------------|
| 4'-Fluoro(1,1'-biphenyl)-4-sulfonyl chloride | 116748-66-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes

severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                              |                                              |
|------------------------------|----------------------------------------------|
| <b>UN-No</b>                 | UN3261                                       |
| <b>Proper Shipping Name</b>  | Corrosive solid, acidic, organic, n.o.s.     |
| <b>Proper technical name</b> | 4'-Fluoro(1,1'-biphenyl)-4-sulfonyl chloride |
| <b>Hazard Class</b>          | 8                                            |
| <b>Packing Group</b>         | III                                          |

### TDG

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>UN-No</b>                | UN3261                                   |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                        |
| <b>Packing Group</b>        | III                                      |

### IATA

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>UN-No</b>                | UN3261                                   |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                        |
| <b>Packing Group</b>        | III                                      |

### IMDG/IMO

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>UN-No</b>                | UN3261                                   |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                        |
| <b>Packing Group</b>        | III                                      |

## 15. Regulatory information

### International Inventories

#### Legend

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**16. Other information**

|                         |                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                 |
| <b>Revision Date</b>    | 16-December-2019                                                                                                                                                                   |
| <b>Print Date</b>       | 16-December-2019                                                                                                                                                                   |
| <b>Revision Summary</b> | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**